Nets joined by label(s): refs auto-assigned
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Sheet: /
File: master_sec.kicad_sch

Title: Master section — goa Console

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POWER ENTRY (7-pin XLR N7MD-LX on the rear panel, two 5 A contacts per rail, 18 AWG to this terminal: +-17 V PSU)

DISTRIBUTION (PTC per wing per rail, 1.6 A hold, 72 V; local branches VA audio, VP phones, VM meters, VL logic)

POWER-ON MUTE: rails > -14 V for ~2 s closes all five relays; either rail low opens them at once

MAIN OUTPUTS: DRV134 -> 1Ω NP in the Sense loop -> mute relay (NO) -> XLR mate, pin 2 hot

HEADPHONES: PHONES dual A10K. OPA1612 + LME49600 composite (+9.9 dB). 2R2 out. PHONESGND

REMOTE METER HEADER (for a meter bridge)

METER L: full-wave rectifier, precision peak / average, follower (TL072). 2 dB ladder. 2 x 10-LED strings

METER R: full-wave rectifier, precision peak / average, follower (TL072). 2 dB ladder. 2 x 10-LED strings

CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs

PFL SWITCHER: series J112 (Delta ED2780 sh.1) -> CRS L/R followers

PFL DETECT (1.5 V per press vs 0.74 V). complementary ramps, rude-solo faster

PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower

AUX MASTER 6: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL

AUX MASTER 5: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL

EXT INPUT (2-track / DAW) and MIX / EXT monitor select (SW latched = EXT)

GROUND CANCEL BUFFER (feeds every re-inverter; summing amps use the raw GCANCEL line)

MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert

MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert

MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB

MONO: inverting sum of POST L+R, re-inverter

MOTHERLOOM (master in the centre: CN1 = left wing, CN2 = right wing; Delta pinout)